

Day 1: Cybersecurity Fundamentals

Worksheet

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Part 1: The CIA Triad - Real World Examples

For each scenario below, identify which CIA principle (Confidentiality, Integrity, Availability) is most affected:

Scenario	CIA Principle	Explanation (1-2 sentences)
Someone reads your private messages without permission	Confidentiality	Unauthorized access to private data violates confidentiality.
A hacker modifies your bank balance	Integrity	Data was altered without authorization, making it inaccurate untrust work.
A DDoS attack makes a website unreachable	Availability	The service is disrupted, So legitimate users cannot access it.
Your password is stolen and used to login	Confidentiality	Stolen credentials allow unauthorized identity impersonation, breaching confidentiality.

Part 2: AAA Security Model

Match the term:

- Verifying who you claim to be Authentication
- Checking if you have permission to access a resource Authorization
- Logging what actions you perform Accounting

a) Checking if you have permission to access a resource b) Verifying who you claim to be c) Logging what actions you perform

Part 3: Types of Attackers

Draw lines to match:

1. Script Kiddie → (1) Uses tools made by others, little skill
 2. Hacktivist → (2) Motivated by political/social causes
 3. Insider → (3) Employee or trusted person
 4. State-Sponsored → (4) Government-backed hackers
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Part 4: Threat vs Vulnerability vs Risk

Fill in the blanks:

- A Vulnerability is a weakness in a system (e.g., unpatched software)
 - A Threat is what could happen (e.g., data theft)
 - A Risk is the likelihood and impact of a threat exploiting a vulnerability
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Part 5: Security Domains - Spot the Domain

Identify which security domain applies:

Situation	Domain
Setting up firewall rules	Network Security
Installing antivirus on laptop	Host /Endpoint Security
Writing secure password code	Application Security
Encrypting customer database	Data Security

Situation	Domain
Lock on server room door	Physical Security

Network Security: It protects network traffic and devices(Firewalls, VPNs, IDS, IPS)

Host / Endpoint Security: Protects individual computers and servers(Antivirus, Updates)

Application Security: Secures software and applications

Data Security: Protects information using encryption and access controls

Physical Security: A lock on the server room door protects physical access to servers and networking equipment. It prevents unauthorized people from entering the room and tampering with the systems.

Part 6: Real Breach Analysis

Pick ONE real breach from the table and describe what went wrong:

Breach	Year	What Happened
Equifax	2017	Unpatched web vulnerability exposed 147M people's data
SolarWinds	2020	Supply chain attack via software update
Colonial Pipeline	2021	Ransomware shut down fuel pipeline
Twitter Bitcoin Scam	2020	Social engineering led to account takeover

Your chosen breach: Equifax (2017)

Root cause (vulnerability): An unpatched Apache Struts web framework vulnerability (CVE-2017-5638). Equifax failed to apply a critical security patch for months, leaving the server exposed.

What CIA principle was violated: Confidentiality — Personal data of 147 million people (SSNs, birth dates, addresses) was accessed by unauthorized attackers.

How it could have been prevented: Timely patch management — applying the available security patch within days of release. Also: network segmentation, intrusion detection systems, and regular vulnerability scanning

Part 7: Responsible Disclosure

Scenario: You found a login bypass on your university's website.

What should you do? (List 3 steps)

Step 1 – Document the vulnerability (don't exploit it): Take screenshots and record the steps to reproduce the issue. Do NOT access other users' accounts or data beyond confirming the bug exists.

Step 2 – Report it to the university's IT/Security team privately: Contact the university's IT helpdesk, CISO, or security team through official channels. Send a detailed but discreet report including the vulnerability type, affected URL, and reproduction steps.

Step 3 – Follow up and allow time to fix: Give them reasonable time (e.g., 30-90 days) to patch the issue before considering public disclosure. If they are unresponsive, escalate through proper channels or use a coordinated disclosure platform.

Part 8: Quick Reflection

What's one cybersecurity concept you want to learn more about? Why?

Web Application Vulnerabilities (specifically SQL Injection and XSS).

The Equifax breach showed how a single unpatched web vulnerability can expose millions of people's data. Understanding how attackers exploit web apps – and how to defend against it – is core to both bug bounty hunting and secure software development. It's directly relevant to real-world penetration testing and building safer applications.

Submission

Submit this completed worksheet as `day1-worksheet.pdf`

Need help: Take a photo or scan - no special format required!
